

Emilie Haynsworth, 1916–1985

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Emilie Virginia Haynsworth was born on June 1, 1916 and died on May 4, 1985, both at home in Sumter, South Carolina. In between she was an inspiration to her family, friends, students, and colleagues. She was born into an old established family in the South; there have been Virginia Emilies or Emilie Virginias in her family for over 200 years. From childhood on, Emilie had a strong and independent mind, so that her intellectual pursuits soon gained her the respect and awe of all her relatives and friends. She was eager to discuss any issue throughout her life. To quote Phil Davis:

She was a strong mixture of the traditional and the unconventional and for years I could not tell beforehand on what side of the line she would locate a given action.

She was greatly loved by her students and not the least for the interest she took in their personal development and in the progress of their research. Her guidance converted many a young graduate student into an able mathematician. She worked successfully with many experts in the field of linear algebra as her papers with P. Davis, M. Fiedler, A. J. Hoffman, M. Marcus, A. Ostrowski, V. Ptak, and others attest. Her mathematical talent evidenced itself in her early years, and by the time she was in junior high school, she was the winner of an annual statewide mathematical contest. She majored in mathematics at Coker College in Hartsville, South Carolina where she re-

ceived an A.B. degree in 1937. She continued her studies at Columbia University, and received an M.A. degree in Mathematics in 1939.

Emilie taught at Olympia High School in Columbia, South Carolina and then at St. Timothy's School in Catonsville, Maryland. During the war years, she worked at Aberdeen Proving Ground in Aberdeen, Maryland, and in 1946, she returned to teaching, this time at the University of Illinois extension in Galesburg.

In 1948, she returned to graduate school at Columbia University. One year later she moved to the University of North Carolina in Chapel Hill where she received her Ph.D. in mathematics in 1952. Alfred T. Brauer was her dissertation adviser (an account of his life is given in [3R]), and the topic of her research was determinantal bounds for diagonally dominant matrices (see [1]). Determinants of diagonally dominant matrices and Gershgorin-type eigenvalue-inclusion regions were a major focus of matrix theory in those days before the implementation of effective numerical procedures on high-speed computers.

This, however, was not the area in which Emilie Haynsworth's work had its greatest impact. If

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad (1)$$

with A nonsingular, then

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

This idea goes back in some form to Gauss, and is of course involved in Gaussian elimination. That the entries of $D - CA^{-1}B$ are minors of M is due to Sylvester [6R] in 1851. If in addition M is square, then Schur [5R] showed in 1917 that

$$\det M = \det A \det(D - CA^{-1}B).$$

In 1968 Emilie Haynsworth named the matrix $D - CA^{-1}B$ the *Schur complement* of A in M , studied its properties as a mathematical object in its own right, and focused our attention on it and its applications. It has turned out to be a very interesting and useful concept in ways unforeseen prior to its christening in 1968.

It is interesting to follow this concept through her work, from 1959 until her last paper, published in 1983. In 1955 Goddard and Schneider [2R]

showed that, if

$$AX = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} B = XB,$$

where A and B are square and X_1 is nonsingular, then the eigenvalues of B are eigenvalues of A , or $\sigma(B) \subseteq \sigma(A)$. (In fact, while this does not appear in [2R], if

$$P = \begin{pmatrix} X_1 & 0 \\ X_2 & I \end{pmatrix},$$

then

$$P^{-1}AP = \begin{pmatrix} B & * \\ 0 & S \end{pmatrix},$$

where S has the Schur complement form $A_{22} - X_2 X_1^{-1} A_{12}$ so that $\sigma(A) = \sigma(B) \cup \sigma(S)$. In 1959 and 1960 Emilie Haynsworth generalized this in [4] and [6] to certain partitioned matrices $[A_{ij}]$, $B = [B_{ij}]$ and $X = \text{diag}(X_i)$ for which $A_{ij} X_j = X_i B_{ij}$ for all i, j by explicitly using matrices S_{ij} of the Schur complement form indicated above.

In 1967 and 1968, [13] and [18], when she and Ostrowski determined the inertia of partitioned hermitian matrices of the form

$$\begin{pmatrix} K_{11} & \cdots & K_{1t} \\ \vdots & \ddots & \\ K_{t1} & & 0 \end{pmatrix}$$

they used as a lemma that

$$\text{In } H = \text{In } H_{11} + \text{In } S_{22},$$

where $\text{In } H$ is the inertia of

$$H = \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^* & H_{22} \end{pmatrix}$$

and S_{22} is the Schur complement of H_{11} in H . And the fact that if H is

positive semidefinite then S_{22} is also, allowed her in [14] (see also [17] and [24]) to show that if H and K are positive definite of the same order, then

$$X^*H^{-1}X + Y^*K^{-1}Y - (X + Y)^*(H + K)^{-1}(X + Y)$$

is positive semidefinite for any rectangular X, Y (generalizing a result of Marcus [4R] for column vectors X, Y). It is in [14] and [17] that the term Schur complement first appeared.

In [14], and with a new proof jointly with Crabtree in [19], she proved a quotient formula for the Schur complement. In [14], and with Burns, Carlson, and Markham in [24] and [25], she used Schur complements (and generalized Schur complements using generalized inverses) to give formulas for inverses of partitioned matrices of the form (1). In connection with generalized inverses, see also the papers [28] and [29] with Wall on group inverses of certain nonnegative matrices.

In 1979, Ando [1R] defined complementable matrices, a generalization of Schur complements in terms of subspaces of C^n instead of partitioned matrices. Haynsworth's last paper [31], jointly with Carlson, extends this work.

Much of Emilie Haynsworth's other work dealt with cones, both "concrete" and abstract. In [15] and, with Hoffman, [20] she dealt with real symmetric matrices H which are copositive, i.e., $x^*Hx \geq 0$ for every non-negative vector x . The following are equivalent for $H \neq 0$: H has the Perron property (the spectral radius is an eigenvalue); H is a positive operator on a closed cone with interior; and H is copositive relative to some self-polar cone in C^n .

With Burns, Fiedler, and Ptak, she studied top-heavy cones [23], algebraic and geometric characterizations of generator matrices of polyhedral cones [26], and extremal operators on cones [27].

Emilie's college teaching career continued in 1951 when she became an instructor at Wilson College in Chambersburg, Pennsylvania. In 1955, she moved to the Washington area where she became a mathematician at the National Bureau of Standards. These were indeed the golden years for matrix theory at the National Bureau of Standards. In attendance during this period were Philip J. Davis, A. J. Hoffman, Marvin Marcus, Morris Newman, Olga Taussky, and John Todd. Alexander Ostrowski and Helmut Wielandt were visitors. For an account of this period see J. Todd [7R].

In 1960, Emilie accepted a position at Auburn University, where she resumed her teaching career. W. V. Parker was head of the department at Auburn, and he interviewed Emilie in Washington prior to her being hired. Emilie used to tell this story: "Dr. Parker came to Washington to interview

me for a job at Auburn. I met him at his hotel. He said, ‘I’ve got this problem in matrix theory...,’ so we worked on his problem until he had to leave. As we were finishing, he told me that I had the job at Auburn.”

From 1960 until her retirement in 1983, Emilie taught with a dedication which honors the teaching profession. She had 18 Ph.D. students, and many more master’s degree students. In 1965, she was named a Research Professor at Auburn, and she was one of the first recipients of this award. Emilie was responsible for a steady stream of distinguished visitors to Auburn, including Professors Phil Davis, Miroslav Fiedler, and Alexander Ostrowski. As a result of her efforts, and those of others, two international matrix theory conferences have been held at Auburn, in 1970 and 1980, both supported by the National Science Foundation. Matrix theory had been represented at Auburn University since the arrival of W. V. Parker in 1950. Emilie carried on this strength until she retired, but well before her retirement she brought in Jim Wall, Bob Grone, and Frank Uhlig to join her and continue the line.

Emilie had little vanity, and did not speak of the honors which came to her. We mention a few of these awards. She was a vice chairman of the Southeastern section of the Mathematical Association of America in 1968–1969, and in 1976–1977, she became the chairman of the Southeastern section. In 1983, Coker College presented her with the Outstanding Alumni Achievement Award during the college’s Alumni Day Celebration.

To those of us who knew Emilie, no recollection would be complete without acknowledging her marvelous sense of humor and her warm humanity. Her positive, witty, kindly, and loving outlook on life was an inspiration to all of us.

A. Ostrowski wrote the following upon her death:

I lost a very good, life-long friend and mathematics an excellent scientist. I remember how on many occasions I had to admire the way in which she found a formulation of absolute originality.

PUBLICATIONS OF EMILIE HAYNSWORTH

1. Bounds for determinants with dominant main diagonal, *Duke Math. J.* 20:199–209, (1953), MR 14, p. 837.
2. Quasistochastic matrices, *Duke Math. J.* 22:15–24 (1955), MR 16, p. 558.
3. Note on bounds for certain determinants, *Duke Math. J.* 24:313–319 (1957), MR 19, p. 628.
4. Applications of a theorem on partitioned matrices, *J. Res. Nat. Bur. Stds.*, 62B:73–78 (1959), MR 22, #318.

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10. A theorem on matrices of 0's and 1's (with M. P. Drazin), *Pacific J. Math.* 13:487–495 (1963), MR 27, #5776.
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19. An identity for the Schur complement of a matrix (with D. E. Crabtree), *Proc. AMS* 22:364–366 (1969), MR 41, #234.
20. Two remarks on copositive matrices (with A. J. Hoffman), *Linear Algebra Appl.* 2:387–392 (1963), MR 40, #1411.
21. Applications of an inequality for the Schur complement, *Proc. AMS* 24:512–516 (1970), MR 41, #241.
22. Reduction of a matrix using properties of the Schur complement, *Linear Algebra Appl.* 3:23–29 (1970), MR 41, #6875.
23. Cones which are topheavy with respect to a norm (with M. Fiedler), *Linear Multilinear Algebra*, 1:203–211 (1973), MR 51, #5630.
24. A generalization of the Schur complement by means of the Moore-Penrose inverse (with D. Carlson and T. Markham), *SIAM J. Appl. Math.*, 26:169–175 (1974), MR 50, #344.
25. Generalized inverse formulas using the Schur complement, (with F. Burns, D. Carlson, and T. Markham), *SIAM J. Appl. Math.* 26:254–259 (1974), MR 48, #8519.
26. Polyhedral cones and positive operators (with F. Burns and M. Fiedler). *Linear Algebra Appl.* 8:547–559 (1974), MR 56, #8598.

27. Extreme operators on polyhedral cones (with M. Fiedler and V. Ptak), *Linear Algebra Appl.* 13:163–172 (1976), MR 52, #13892.
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30. Review of *Circulant Matrices*, by Philip Davis, *Linear Algebra Appl.* 43:243–244 (1982).
31. Complementable and almost definite matrices (with D. Carlson), *Linear Algebra Appl.* 52/53:157–176 (1983), MR 84i, #15015.

PH.D. STUDENTS OF EMILIE HAYNSWORTH (ALL AT AUBURN UNIVERSITY)

Marjorie H. Fitzpatrick	Kronecker and other special products of matrices, June 1964.
Donald H. Clanton	Some algorithms for the calculation of the characteristic roots and vectors of a normalizable matrix, August 1964.
Edmond D. Dixon	Matric polynomials which are higher commutators, June 1965.
James N. Issos	The field of values of non-negative irreducible matrices, June 1966.
Rex C. Jones	Composite tridiagonal matrices, August 1966.
James J. Johnson	Bounds on the permanent of a dominant diagonal matrix, June 1967.
Thomas L. Markham	On oscillatory matrices, June 1967.
Frank L. Salzmann, III	Copositive matrices and positive operators on cones, August 1970.
Luck J. Watford Jr.	M -matrices with respect to a cone, June 1971.
Clyde W. Davis Jr.	The Ljapunov and Stein transformation and related results, August 1971.
Isham F. Burns	Topics in cones and positive operators, June 1973.
Teh-huey Chuang	J -matrices and quasi-partial isometries, August 1973.
Walter J. Hattaway	The Chipman inverse, March 1974.
Benjamin F. Caldwell	Methods of construction of matrix-valued functions with applications to the generalized inverse, August 1975.
Michael W. Poole	Structure properties of polyhedral cones, August 1975.

- Ronald L. Smith Generalized circulant and block centrosymmetric matrices, August 1976.
- Gregory A. Johnson A generalization of N -matrices, December 1980.
- Cornelius L. Bell, Jr. Results in the theory of circulant and other normal matrices, August 1981.

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